The document "Video Encoding with PyAV" details a comprehensive study on video encoding using PyAV, focusing on different configurations for **H.264** and **H.265** codecs, as well as experiments with the **AV1** codec. Below are the key findings and methodologies:

1. Encoding Configurations

Three primary encoding techniques were tested:

Only I-frames: Every frame is an I-frame (keyframe), ensuring full independence but resulting in larger file sizes.

Single I-frame and rest P-frames: One I-frame followed by P-frames (predictive frames), offering better compression than all I-frames.

Single I-frame with interleaved P and B-frames: Incorporates B-frames (bi-directional frames) for higher compression efficiency.

2. Codec Comparison (H.264 vs. H.265)

H.265 (HEVC) consistently outperformed H.264 in compression efficiency, producing smaller file sizes for the same CRF (Constant Rate Factor) values.

For example:

Only I-frames: H.265 files were ~40-50% smaller than H.264 at the same CRF.

Interleaved P and B-frames: H.265 achieved even greater savings, especially at higher CRF values (e.g., crf = 40).

3. Impact of CRF Values

Lower CRF (e.g., 18): Higher quality but larger file sizes.

Higher CRF (e.g., 40): Lower quality but significantly smaller files.

Optimal CRF Range: 18-23 provided a good balance between quality and compression.

4. GOP (Group of Pictures) Size

A GOP size of 15 was standard, but H.265 sometimes extended it to 30 for better compression.

Larger GOP sizes (e.g., 150, 250) were tested for static camera videos, showing further size reductions.

5. AV1 Codec Experiments

Golden Frames: Enabled with specific parameters (enable-golden-frame=1, hierarchical-levels=4, etc.), these frames improved compression efficiency.

Results: AV1 with golden frames produced smaller files compared to traditional H.264/H.265 configurations, especially at higher CRF values.

6. Other Findings

QP (Quantization Parameter) vs. CRF: QP showed similar trends to CRF but with slight variations in file sizes.

Static vs. Dynamic Videos: Static videos (e.g., surveillance footage) achieved better compression with larger GOP sizes and interleaved B-frames.

7. Verification Tools

ffprobe: Used to confirm frame types (I, P, B) in encoded videos.

Golden Frame Detection: Verified via metadata and frame patterns in AV1-encoded videos.